

# Abdullah Khaled

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## Education

- Wakeland High School**, High School Diploma – Frisco, TX Aug 2023 – May 2027
- Unweighted GPA: 4.0
  - 18 AP courses by graduation (completed and planned)
- Frisco ISD Career and Technical Education Center**, Project Lead The Way – Frisco, TX Aug 2023 – May 2027
- Introduction to Engineering Design, Principles of Engineering, Digital Electronics, Engineering Design and Development
  - PLTW Engineering EPortfolio: [akhaled247.github.io/pltw-engineering](https://akhaled247.github.io/pltw-engineering)

## Technical Experience

- Programming Lead, Chief Technical Advisor**, ITKAN Robotics (FRC 9128) – Frisco, TX June 2024 – present
- Developed components of competition robot software stack integrating localization, path planning, telemetry, and mechanism control in Java
  - Built publish-subscribe telemetry and motor logging framework for real-time robot diagnostics and performance analysis
  - Led development of robot state estimation and localization system using sensor fusion of AprilTag vision and swerve-drive odometry, achieving <2 in positional error
  - Implemented finite-state machine control architecture with Dijkstra-based path planning using time-weighted graph
  - Used WPILib Java mechanism simulation to optimize telescoping arm geometry and motion constraints
  - Developed self-paced robotics programming curriculum and training website used by 15+ FRC and FTC students ([mantik.netlify.app](https://mantik.netlify.app))
  - Advanced team to Texas State Championship playoffs (89 teams)
  - Robot codebase and documentation: [akhaled247.github.io/first/reefscape](https://akhaled247.github.io/first/reefscape)
- Programming Mentor**, ITKAN Robotics of ICF (FTC 26357, 28931) – Frisco, TX Sept 2024 – Feb 2026
- Implemented Kalman-filtered state estimation combining AprilTag vision and wheel odometry for real-time robot localization and automated turret targeting
  - Mentored all-rookie FTC team through full engineering design cycle from prototyping to competition
  - Mentor refugee students in FLL workshops and lead weekly summer training for FTC and FRC students using self-developed OARobotics curriculum ([akhaled247.github.io/personal/oarobotics](https://akhaled247.github.io/personal/oarobotics))
- Co-Founder, President**, Wakeland High School Wolverine Robotics (FTC 33791) – Frisco, TX Aug 2024 – present
- Co-founded Wakeland robotics club, growing membership from 0 to 50+ students in the first year
  - Secured funding and constructed competition-ready robot in under 2 weeks
  - Teach CAD software (Onshape) and Java programming through monthly technical workshops
  - Manage club operations including mission development, curriculum design, and budget management
- Math and Reading Instructor**, Gideon Math and Reading Center – Frisco, TX June 2024 – present
- Instruct K-11th grade students in mathematics (Algebra, Geometry), reading, and grammar
  - Analyze student performance data to identify learning gaps and develop personalized plans
  - Evaluate reading material difficulty and adjust instruction to match target comprehension levels

## Projects

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### OralVision – The Conrad Challenge

Oct 2025 – present

- Designed oral disease detection system using HIPAA-compliant federated learning with TensorFlow Federated, enabling distributed model training without sharing patient data
- Used Onshape 3D CAD software to design and 3D print the device, reducing manufacturing costs

### WatchFall – Samsung Solve for Tomorrow

Sept 2025 – present

- Designed fall-detection device using Raspberry Pi Pico with on-device TensorFlow Lite inference
- Developed lightweight ML model for real-time motion tracking and fall detection with buzzer alert system
- Engineered hybrid dataset combining public and custom annotated data, reducing false positive rate

## Skills

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**Programming:** Python, Java, HTML, CSS, JavaScript

**CAD/CAM:** Fusion 360, Onshape

**Web Development:** Jekyll

**Computer Vision:** OpenCV, SolvePnP, TensorFlow

**Electronics:** NI Multisim, KiCAD, EasyEDA, Soldering

**Data & Statistical Analysis:** R, Google Sheets

**Productivity Tools:** Google Workspace, Microsoft Office

## Leadership & Activities

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**Vice President,** Math National Honor Society – Wakeland High School, Frisco, TX

Sept 2023 – present

- Led organization as Vice President (2024-25) and Treasurer (2023-24), managing budget for 100+ members
- Designed Math Lab Program website using Google Sites and coordinated peer tutoring, attaining over 20% member participation
- Organized annual middle school math competition with 100+ participants, 3 participating schools, and 35+ volunteers

**Secretary, Co-Founder,** Muslim Student Association – Wakeland High School, Frisco, TX

Sept 2024 – present

- Co-founded MSA as Secretary, establishing organizational structure and creating digital promotional media
- Negotiated with school administration to establish dedicated prayer room, coordinating facility setup and policies

**Lead Math and English Instructor,** Ma'ruf Project Taleem – Dallas, TX

Sept 2023 – present

- Teach Math and English to refugee elementary and middle school students, developing differentiated curriculum
- Provide mentorship in academic skills, time management, and social integration for students transitioning to American schools

## Honors & Awards

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- (2026) OralVision: Conrad Challenge Innovator Award & Alternate Finalist, Diamond Challenge Semifinalist
- (2026) FTC 33791: Event Winner, Control Award Winner (FiT-North Dallas Semi-Regional Championship, 25 teams), Control Award Winner (FiT-North U-League Tournament)
- (2025) FRC 9128: FIRST® Leadership Award Semifinalist, Event Winner (FIRST® in Texas Victoria, 25 teams); Autonomous Award (FiT Plano, 31 teams); Creativity Award (FiT Amarillo, 36 teams)
- (2025) FTC 26357: Finalist Alliance Captain (FiT-North McKinney Qualifier, 19 teams)
- (2025) Collegeboard: AP Scholar with Distinction, National Recognition Program: School Recognition Award